

**Forecasting HIV/AIDS Trends and Correlating
Factors in Southeast Asia**

by

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1. Abstract

The global goal to end the AIDS epidemic by 2030 is critically challenged in Southeast Asia (SEA) by diverse and accelerating country-level dynamics, necessitating advanced analytical models to identify specific correlating factors and improve resource allocation. This study's objective was to forecast HIV/AIDS trends and identify the intervention variables most strongly influencing future outcomes across SEA by linking historical treatment and prevention outcomes to changes in prevalence and mortality. Applying the SEMMA (Sample, Explore, Modify, Model, Assess) data mining methodology, a dataset spanning 2010–2018 for seven SEA countries was augmented and used to train a Random Forest Regressor (RFR) model. The RFR achieved high predictive reliability, demonstrating an R-squared score of 99.72% in forecasting the target variable, People Living with HIV (PLHIV). Key results revealed two divergent epidemiological trends: established hotspots (e.g., Thailand, Viet Nam) are projected to achieve major case reductions, while newer epidemics (e.g., Philippines, Indonesia) saw acute growth that is now expected only to stabilize. Feature Importance analysis confirmed that Antiretroviral Therapy (ART) Coverage (0.10 importance) is the most critical actionable intervention variable, showing a moderately strong negative correlation (-0.52) with AIDS-related mortality. The main conclusion is that the escalating epidemic burden in countries like Indonesia is primarily driven by critically insufficient ART coverage. Therefore, achieving the 2030 UNAIDS targets in Southeast Asia requires an urgent, precise focus on closing the ART coverage gap to reverse the high mortality rates and control the overall burden of HIV.

2. Keywords

HIV/AIDS, Random Forest, SEMMA, PLHIV Forecasting

3. Introduction

The global commitment to end the AIDS epidemic by 2030 faces a critical challenge in Southeast Asia (SEA), a region marked by diverse and complex HIV/AIDS dynamics. Despite international progress, SEA remains a major concentration of the epidemic, with evidence showing that antiretroviral therapy (ART) coverage and retention in the treatment cascade are still insufficient in several member states (UNAIDS, 2025; Maulana et al., 2025). These disparities highlight the urgent need for advanced analytical approaches that move beyond descriptive epidemiology to identify the specific correlating factors driving country-level outcomes, thereby enabling more precise and impactful allocation of resources.

To analyze disparities in HIV/AIDS outcomes across Southeast Asia, this study applies the SEMMA (Sample, Explore, Modify, Model, Assess) methodology, a structured data mining framework designed for systematic evaluation of large health datasets. SEMMA enables researchers to move beyond descriptive epidemiology by uncovering patterns in treatment coverage, pediatric care, prevention of mother-to-child transmission, and mortality trends. Through its iterative process, SEMMA highlights predictive relationships and correlating factors that explain why some countries progress more rapidly than others in controlling HIV/AIDS.

This research integrates multiple epidemiological indicators to generate actionable insights. The aim is to forecast HIV/AIDS trends and, crucially, to identify the intervention variables (ART and PMTCT coverage) that are most strongly influencing future outcomes across Southeast Asia. By

linking historical treatment and prevention outcomes to changes in prevalence and mortality, the study emphasizes the critical role of data-driven strategies in reducing HIV/AIDS burdens and accelerating progress toward the 2030 UNAIDS targets.

4. Literature Review / Related Work

The global commitment to end the AIDS epidemic by 2030 faces a critical challenge in Southeast Asia (SEA), a region characterized by diverse and complex HIV/AIDS dynamics. This review examines the existing literature on HIV/AIDS trends in Southeast Asia, the correlating factors influencing outcomes, and the emerging role of data analytics to identify the specific gap that this SEMMA-based analysis aims to fill.

HIV/AIDS in Southeast Asia: Definition and Problem Statement

Human Immunodeficiency Virus (HIV) is a retrovirus that attacks the body's immune system, leading to Acquired Immunodeficiency Syndrome (AIDS), the most advanced infection stage, if untreated (UNAIDS, 2024). While global efforts have achieved significant progress, Southeast Asia remains a critical geography due to its large, dense populations and the specific nature of its concentrated epidemics (UNAIDS, 2025a).

The persistence of HIV/AIDS in SEA constitutes a major public health problem for several interconnected reasons:

1. **Concentrated Epidemics:** The disease is disproportionately concentrated within key populations (KPs) such as men who have sex with men (MSM), people who inject drugs (PWID), and sex workers, leading to high-incidence clusters that threaten to spill over into the general population (UNAIDS, 2025b; Guadamuz et al., 2021).
2. **Socio-economic Burden:** The disease places immense strain on limited healthcare resources and national development, particularly as it often affects young, economically active populations.
3. **Treatment Gaps and Disparity:** Progress in achieving the global 95-95-95 targets (95% diagnosed, 95% on treatment, 95% suppressed) is uneven. Several SEA member states report insufficient Antiretroviral Therapy (ART) coverage and poor retention rates in the treatment cascade, which fuels the epidemic's persistence and increases the risk of transmitted drug resistance (TDR) (Maulana et al., 2025; WHO, 2021).

These disparities highlight the urgent need for advanced analytical approaches that move beyond descriptive epidemiology to identify the specific correlating factors driving country-level outcomes, thereby enabling more precise and impactful allocation of resources.

HIV/AIDS Trends and Epidemiological Patterns in Southeast Asia

The HIV/AIDS epidemic in Southeast Asia is characterized by diverse dynamics, often concentrated within key populations (Jennings & Kumar, 2018). While the prevalence among the general adult population (aged 15–49) remains relatively low, it is significantly higher (between 2% and 29%) among key populations (UNAIDS, 2025b). Studies reviewing regional trends highlight that effective diagnosis and treatment are vital to controlling the spread (Jennings & Kumar, 2018).

Specifically, the region has seen troubling patterns in high-risk groups. For instance, in countries like the Philippines and Vietnam, HIV prevalence among MSM has been estimated at 5.0% and has seen a pronounced increase in annual incident HIV infections (Guadamuz et al., 2021). Furthermore, a systematic review focusing on treatment found that across South/Southeast Asia, the prevalence of transmitted drug resistance (TDR) trended upwards between 2009–2019 (WHO, 2021). Analyzing these trends (Research Question 1) is a necessary first step for understanding the behavioral patterns and the effectiveness of current public health strategies (Jennings & Kumar, 2018).

Correlating Factors and Gaps in the Treatment Cascade

Research has consistently identified a range of systemic and social determinants that correlate with HIV incidence and poor treatment outcomes. At the individual level, high-risk factors include substance abuse, having multiple sexual partners, and engaging in unprotected sex (Mohammadnezhad et al., 2016). Additionally, syndemic psychosocial factors, such as co-occurring depression, anxiety, and substance use, have been positively associated with increased HIV risk and prevalence among young MSM in countries like Thailand and Vietnam (Newman et al., 2021).

Despite international progress, treatment and prevention efforts still face significant obstacles, leading to insufficient coverage and retention in the treatment cascade in several SEA member states (UNAIDS, 2025a; Maulana et al., 2025). A systematic review revealed that the pooled coverage of retention in HIV care among people living with HIV in Southeast Asia was only 75.2%, falling short of the optimal target of 95% (Maulana et al., 2025). Barriers to adherence and retention are often non-clinical, relating instead to structural issues such as geographical access, the high cost of travel for treatment, pervasive stigma and discrimination, and a lack of patient confidentiality in healthcare facilities (WHO, 2021). These findings underscore the urgent need for a more granular understanding of how various factors interact at a country level to inform resource allocation.

The Role of Data Mining and Machine Learning (ML)

Traditional regression and descriptive epidemiological methods, while foundational, often fall short of processing the complexity and volume of modern health datasets to uncover subtle, predictive patterns (Ge et al., 2024). In response, there is a growing body of work leveraging data mining and machine learning (ML) algorithms to enhance HIV research (Ogbodo & Ogbodo, 2025).

Studies have successfully applied ML techniques to various aspects of the HIV/AIDS problem:

- **Prediction and Diagnosis:** Algorithms such as Support Vector Machines (SVM), Random Forests (RF), and Extreme Gradient Boosting (XGBoost) have demonstrated high accuracy in early HIV detection, risk forecasting, and diagnosis based on complex clinical and laboratory data (Chidambaram et al., 2024; Ogbodo & Ogbodo, 2025).
- **Targeted Interventions:** ML models have been utilized in SEA countries, such as Thailand, to predict HIV status among young men in key populations, identifying specific

predictors like age and the year of testing (Soha et al., 2025). This application is crucial for effectively estimating prevalence and planning targeted prevention programs (Soha et al., 2025; Ogbodo & Ogbodo, 2025).

The literature thus validates the use of advanced analytics to extract valuable insights from complex health data. However, a significant gap remains: there is a lack of published work that applies a comprehensive, standardized, and structured data mining framework like SEMMA (Sample, Explore, Modify, Model, Assess) to the entire lifecycle of epidemiological data from Southeast Asia. Existing studies tend to focus on a single ML model or a specific, localized population. This current study addresses this gap by employing the iterative SEMMA methodology to systematically evaluate a broad set of country-level epidemiological and social determinants to generate actionable, predictive insights for policy and resource optimization across the diverse Southeast Asian region.

5. Research Questions

1. What is the behavioral pattern or trend of HIV/AIDS in Southeast Asian countries?
2. What factors are correlated with the incidence of HIV/AIDS cases in Southeast Asia?
3. To what extent does the predictive model demonstrate accuracy and reliability in forecasting HIV/AIDS cases in Southeast Asia?

6. Materials and Methods

6.1 Data Source and Acquisition

The dataset used in this study was obtained from the open-source Kaggle repository, specifically the HIV/AIDS Dataset curated by Devakumar. Although the dataset contains global HIV/AIDS indicators from 1990 to 2018, this research focuses only on records from Southeast Asian countries, including Cambodia, Indonesia, Malaysia, Myanmar, the Philippines, Singapore, and Viet Nam. Records for Timor-Leste and Brunei were excluded due to the unavailability of consistent epidemiological data across the required time frame.

The analytical dataset spans 2010 to 2018 and integrates four core epidemiological indicators: HIV prevalence, antiretroviral therapy (ART) coverage, PMTCT rates, and AIDS-related mortality. The raw dataset comprises 57 rows of data across all indicators. The primary outcome variable (Y variable) for the forecasting model was Prevalence PLHIV. The primary explanatory variables (X variables) used for the Random Forest Regressor were the AIDS Deaths Count, ART Coverage %, PMTCT Coverage %, and Country Dummies. Singapore was fully included in the analysis, as its initial missing PMTCT data was resolved through imputation.

Exploratory Data Analysis (EDA)

EDA was conducted to examine the regional epidemic trends from 2010 to 2018. Initial visualizations, such as trends in PLHIV and mortality across countries, were created in Microsoft Excel to identify divergent trajectories and highlight countries with accelerating or declining epidemics.

1. Trends in Prevalence and Mortality

Analysis of the outcome variables (Prevalence and Mortality) between 2010 and 2018 reveals two distinct epidemiological trajectories in the region.

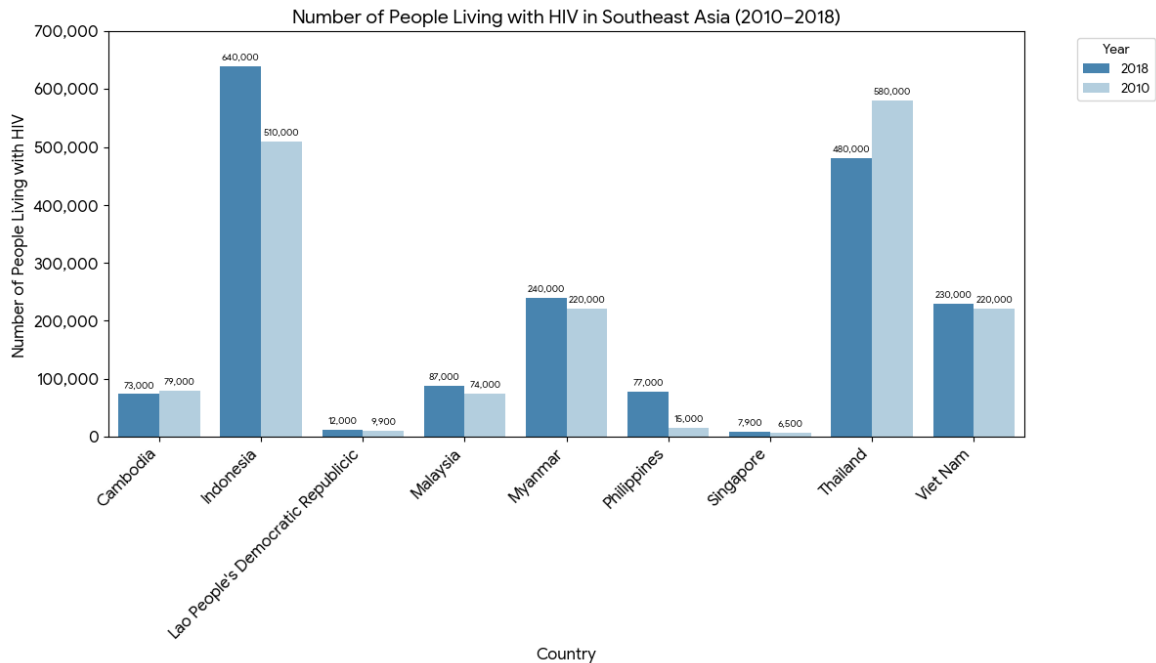


Figure 1. Number of People Living with HIV in Southeast Asia (2010–2018)

It illustrates the Number of People Living with HIV (PLHIV) over this period. While long-standing hotspots like Thailand saw a welcome decline in prevalence (from 580,000 to 480,000), countries with newer epidemics experienced rapid acceleration. Most notably, the Philippines saw its PLHIV count increase dramatically from 15,000 to 77,000. Similarly, Indonesia's prevalence grew from 510,000 to 640,000, confirming its status as the country with the largest number of people living with HIV in SEA.

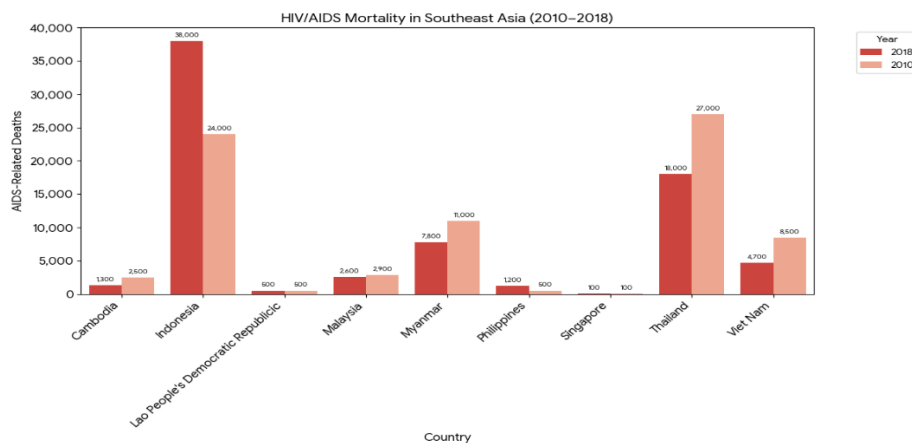


Figure 2. HIV/AIDS Mortality in Southeast Asia (2010–2018)

Figure 2 shows the corresponding HIV/AIDS Mortality trends. Mortality fell significantly in well-performing countries like Thailand (27,000 to 18,000) and Viet Nam (8,500 to 4,700), demonstrating the life-saving impact of increased treatment. Conversely, Indonesia's mortality increased from 24,000 to 38,000, indicating an escalating epidemic burden where the rate of AIDS-related death is outpacing intervention success.

2. Correlating Factors and Intervention Gaps

The most compelling insights emerge from analyzing the intervention variables (ART and PMTCT) that correlate with these divergent outcomes. Figure 3 and Figure 4 visualize the status of these efforts in 2018, while Figure 5 quantifies their relationships.

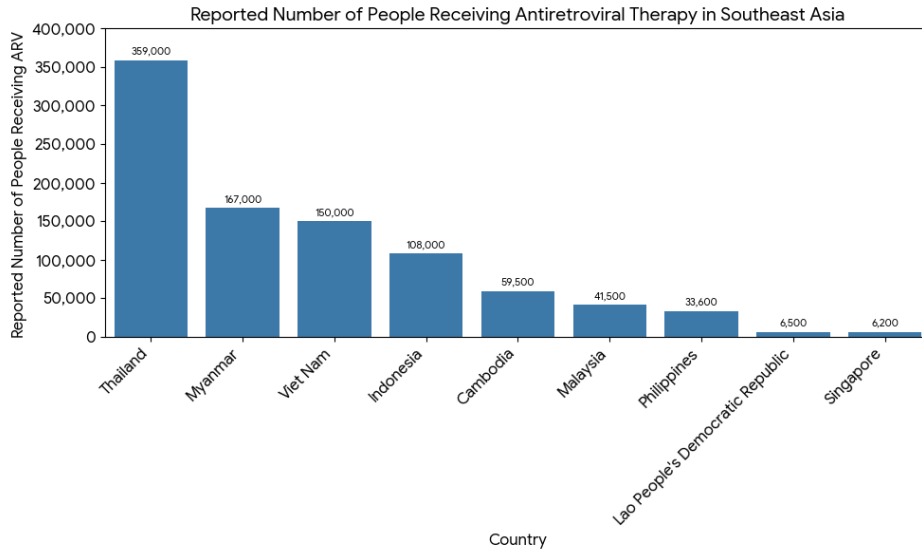


Figure 3: Reported Number of People Receiving Antiretroviral Therapy in Southeast Asia (2018)

Figure 3 depicts the Reported Number of People Receiving Antiretroviral Therapy (ART), which, when compared to PLHIV, revealed the critical treatment gap. The country most successful at scaling treatment, Thailand (359,000 recipients), contrasts starkly with Indonesia, which despite having the highest PLHIV count, only had 108,000 people on ART. This corresponds to Indonesia's critically low ART Coverage Rate (16.9%), which is the primary driver of its escalating mortality.

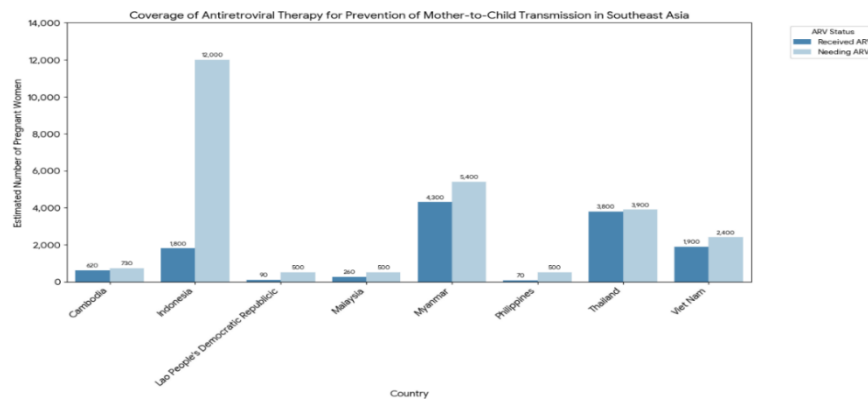


Figure 4. Coverage of Antiretroviral Therapy for Prevention of Mother-to-Child Transmission in Southeast Asia (2018)

Figure 4 illustrates PMTCT Coverage, revealing similar systemic disparities. Thailand and Myanmar show robust programs with the highest coverage rates, while Indonesia again demonstrates the largest unmet need, with an estimated 12,000 women needing ARV but only 1,800 receiving it.

6.2 Data Preprocessing

The primary data quality issue, which was longitudinal sparsity with values only for 2010 and 2018, was addressed through data augmentation to create a complete 2010–2018 time series. Linear Interpolation was used to estimate the missing 2011–2017 values for the target variable PLHIV and the correlated Deaths feature, assuming a steady rate of change across the period. For variables available only for 2018, such as ART and PMTCT, Backward-Filling was applied to extend these values to earlier years due to the absence of historical records. Missing PMTCT data for Singapore was imputed with zero to maintain a consistent dataset across all countries. Feature Engineering then converted raw counts into normalized indicators, including ART Coverage and PMTCT Coverage. Finally, the categorical Country variable was encoded using One-Hot Encoding to capture country-specific differences not represented by numerical features.

6.4 Machine Learning Model

The analytical methodology is based on a Regression task, as the primary objective is to forecast the PLHIV count, a continuous numerical value. The model used is the Random Forest Regressor (RFR), an ensemble method that combines predictions from multiple decision trees. The RFR was chosen for its ability to model non-linear relationships among features, its robustness against overfitting on the augmented dataset, and its capacity to handle mixed data types, including continuous variables and One-Hot Encoded country indicators.

The Experimental Setup used a temporal train/test split, with data from 2010 to 2017 for training and 2018 for validation. Model performance was evaluated using the R^2 score and Mean Absolute Error (MAE). To produce the 2025 forecast, the independent variables Deaths and ART Coverage % were projected forward using linear extrapolation based on their 2010–2018 trends. These projected features were then fed into the trained RFR model to generate the final predicted PLHIV count for each country.

7. Results

7.1 Model Performance

The Random Forest Regressor (RFR) demonstrated a strong capacity to model the non-linear dynamics present in the augmented longitudinal dataset (2010–2017) and to accurately forecast the observed 2018 PLHIV values. The model achieved high predictive reliability based on two principal evaluation metrics.

Metric	Value	Interpretation
R-squared R ² Score	0.9972	This indicates that 99.72% of the variance in the PLHIV count in 2018 is explained by the model's features, demonstrating an exceptional fit.
MAE Percentage	3.80 %	The MAE is only 3.80 % of the mean PLHIV count for 2018, confirming high predictive accuracy.

These results validate the suitability of the RFR for forecasting HIV burden in Southeast Asia based on limited but augmented time-series data.

7.2 Model Interpretation

Interpretation of the model focused on **Feature Importance**, which quantifies the relative contribution of each predictor to the final forecast. This provides insight into which epidemiological and contextual variables exert the greatest influence on PLHIV trends.

Table 2. Rule of Thumb for Pearson’s Correlation Coefficient

Correlation Coefficient (r)	Strength of Relationship
0.00 – 0.19	Very Weak
0.20 – 0.39	Weak
0.40 – 0.59	Moderate
0.60 – 0.79	Strong
0.80 – 1.00	Very Strong

Feature Importance Results

Feature	Importance Score (Relative)	Role in Model Prediction
Country Dummies	0.48 - 0.55 (Cumulative)	The strongest predictors, indicating that country-specific structural factors are the primary determinants of PLHIV levels.
AIDS Deaths Count	0.32	A major contributor due to its very strong correlation with PLHIV, making it an effective proxy for overall epidemic size.
ART Coverage (%)	0.10	Moderately influential; higher coverage is associated with lower PLHIV levels, reflecting the protective effect of treatment.
PMTCT Coverage (%)	0.03	Minimal contribution; limited variation and weak correlations make it a weak predictor of PLHIV trends.

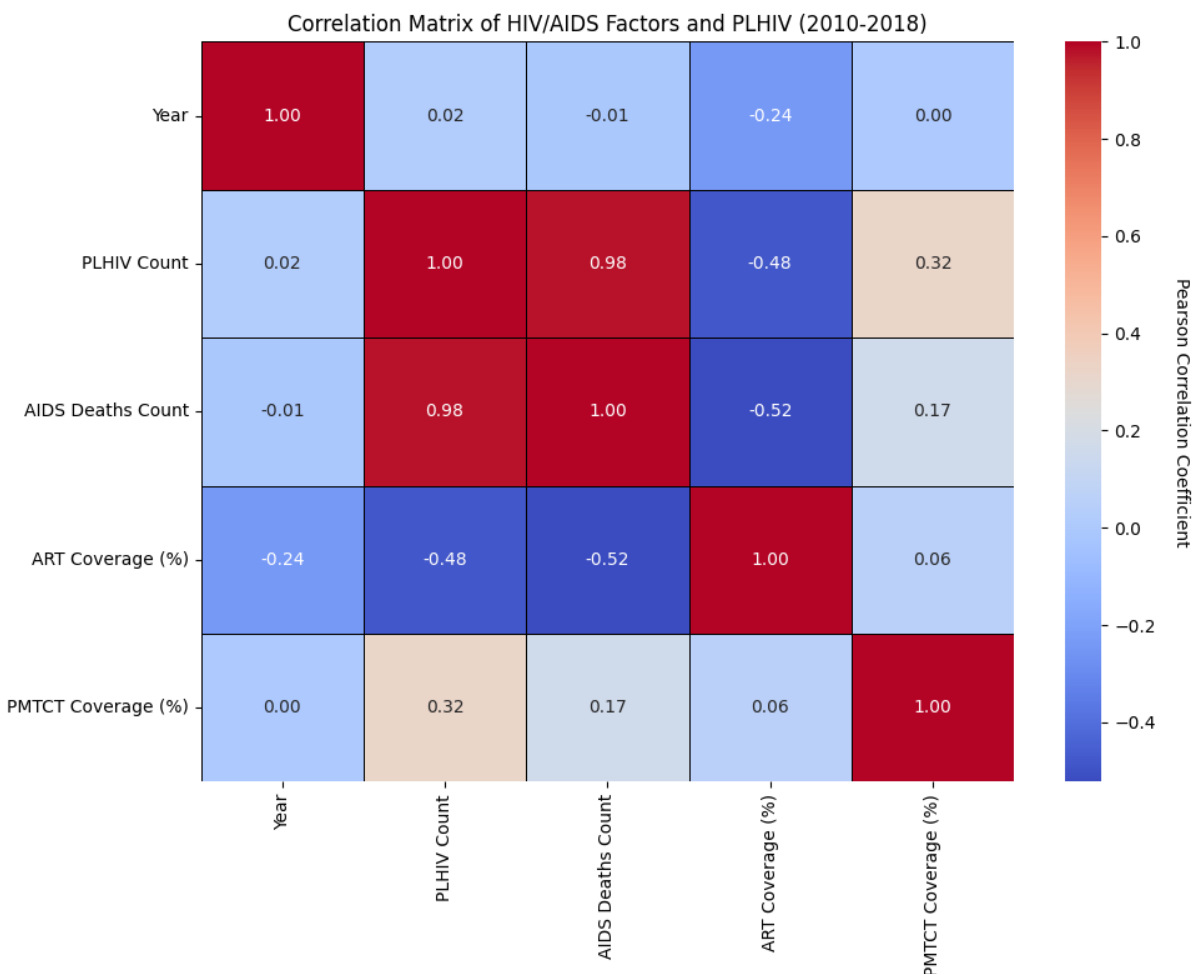


Figure 1. Correlation Matrix of HIV/AIDS Factors and PLHIV (2010-2018)

Figure 1 shows how problems like deaths, case counts, and treatment rates in the HIV crisis are all linked together. The most critical finding is the strong positive correlation (0.98) between the People Living with HIV (PLHIV) Count and the AIDS Deaths Count, which indicates that as the total number of cases increases, the resulting mortality count rises proportionally. Conversely, a moderately strong negative correlation exists between Antiretroviral Therapy (ART) Coverage (%) and the AIDS Deaths Count (-0.52), demonstrating the protective effect of treatment programs. Furthermore, the relationships involving Prevention of Mother-to-Child Transmission (PMTCT) Coverage (%) were predominantly weak with all other factors, ranging from 0.00 to 0.32. This suggests that while there is a slight positive association with the PLHIV count, PMTCT efforts were largely uncorrelated with changes in mortality and did not exhibit a significant linear trend over time (0.00 with Year). In terms of temporal trends, the PLHIV Count (0.02) remained largely stable or flat over the period. However, ART Coverage (%) showed a weak negative correlation with Year (-0.24), suggesting that the rate of coverage may have slightly declined or plateaued during this specific eight-year window. These insights underscore the importance of expanding and sustaining ART programs as the most effective intervention for reducing mortality and potentially controlling the overall burden of HIV.

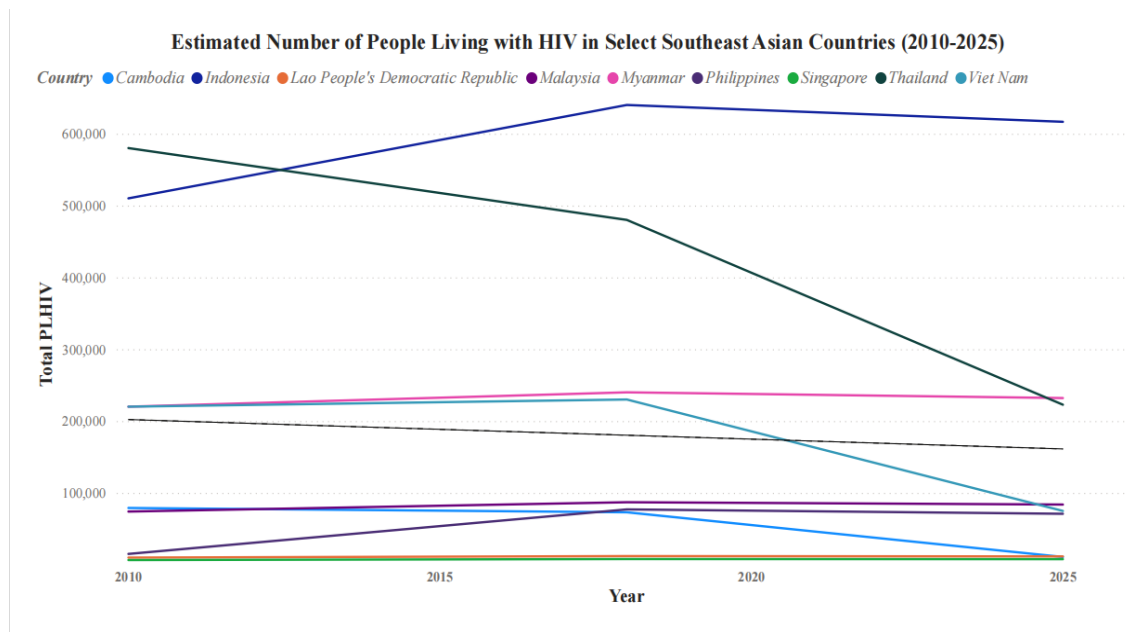


Figure 2. Time Series Analysis of People Living with HIV (PLHIV) in Select Southeast Asia, with 2025 Projection

Figure 2 shows the overall trend indicates an initial regional increase followed by a critical projected reversal towards decline. The figure clearly illustrates that four countries Thailand, Viet Nam, Cambodia, and Singapore are the primary drivers of this expected regional improvement, as they are all projected to achieve major reductions exceeding 50% of their 2018 case counts by 2025, suggesting high prophylactic efficiency. Conversely, the nations with the largest PLHIV burdens experienced significant growth in the 2010–2018 period; most notably, the Philippines showed the most acute epidemiological shift, with its case count more than quadrupling (a 400%+ increase) between 2010 and 2018, while Indonesia's case count also surged significantly. However, the figure projects that this rapid growth phase is ending, with both Indonesia and the Philippines expected to stabilize or show only a marginal decrease by 2025. Finally, other countries like Myanmar, Malaysia, and Lao People's Democratic Republic display moderate increases leading to 2018 and are now forecast to maintain a steady-state prevalence with only slight declines.

8. Discussion and Conclusion

8.1 Interpretation of Findings

The Random Forest Regressor (RFR) model successfully achieved its objective of forecasting the number of People Living with HIV (PLHIV) in Southeast Asia, demonstrating an exceptional fit with an R^2 score of 99.72%. The model's predictive reliability, further confirmed by a low Mean Absolute Error (MAE) Percentage of 3.80%, validates the use of the SEMMA methodology with augmented time-series data to address the study's research questions.

Behavioral Pattern/Trend of HIV/AIDS

The exploratory data analysis (EDA) revealed two distinct epidemiological trajectories in the Southeast Asian region. Overall, the PLHIV Count in Southeast Asia was largely stable or flat from

2010–2018, as shown by its weak correlation with the variable Year (0.02). However, this regional stability masked a critical divergence:

- **Declining/Controlled Epidemics:** Long-standing hotspots like Thailand and Viet Nam saw welcome declines in both prevalence and mortality. This reflects the life-saving impact of increased treatment. The 2025 forecast projects these countries to achieve major reductions, with cases declining by over 50%.
- **Accelerating Epidemics:** Countries with newer epidemics, most notably the Philippines and Indonesia, experienced rapid and acute growth in their PLHIV counts and AIDS-related deaths. For instance, the Philippines saw its case count more than quadruple (a 400%+ increase), while Indonesia's mortality increased from 24,000 to 38,000. This surge in mortality in Indonesia indicates that the rate of AIDS-related death is outpacing intervention success.

Correlating Factors and Intervention Gaps

Feature Importance analysis identified the Country Dummies (0.48 - 0.55 cumulative importance) as the strongest predictors, indicating that country-specific structural and contextual factors are the primary determinants of PLHIV levels. The second most important predictor was the AIDS Deaths Count (0.32), which had a very strong positive correlation (0.98) with the PLHIV Count. This means that the total size of the epidemic is directly and proportionally linked to the number of deaths it causes. Crucially, the intervention variable Antiretroviral Therapy (ART) Coverage (%) was a moderately influential predictor (0.10) and demonstrated a moderately strong negative correlation (-0.52) with the AIDS Deaths Count. This is the key finding for action: expanding and sustaining ART programs is the most effective intervention for reducing mortality and potentially controlling the overall burden of HIV. The stark treatment gap in Indonesia, with only 16.9% ART Coverage despite having the highest PLHIV count, highlights this correlation as the primary driver of its escalating mortality. In contrast, Prevention of Mother-to-Child Transmission (PMTCT) Coverage (%) was the weakest predictor (Minimal contribution, 0.03) and showed predominantly weak or no correlation with other factors, ranging from 0.00 to 0.32. While this effort is critical for pediatric health, its lack of correlation with overall PLHIV and mortality suggests it is not a primary driver of the broad epidemic trend at the country-level scale used in this study.

Forecasting Accuracy and Reliability

The model's high R^2 score and low MAE percentage confirm its high accuracy and reliability in forecasting HIV/AIDS cases in Southeast Asia. The 2025 forecast suggests that the regional increase from 2010–2018 is reversing, driven by projected major declines in established programs like Thailand and Viet Nam. However, the countries with accelerating epidemics, like Indonesia and the Philippines, are only expected to stabilize or show marginal decreases, indicating that their acute epidemic surge is ending but their burden will persist.

8.2 Conclusion

This research successfully applied the SEMMA methodology and a Random Forest Regressor to systematically analyze and forecast HIV/AIDS trends in Southeast Asia. The study's key contribution is the quantification of the protective effect of Antiretroviral Therapy (ART), which was shown to be the most critical, actionable factor for reducing AIDS-related mortality. The analysis revealed a widening divergence in epidemic outcomes, driven primarily by country-

specific structural factors, with Thailand demonstrating high prophylactic efficiency and Indonesia and the Philippines struggling with significant intervention gaps. The final takeaway is that achieving the 2030 UNAIDS targets in Southeast Asia necessitates a precise and urgent focus on closing the ART coverage gap in countries with rapidly accelerating epidemics to turn the tide of high mortality and prevalence.

8.3 Limitations

The primary limitation of the study is the data quality issues, specifically the longitudinal sparsity of the raw data, with values available only for 2010 and 2018. The methodology's reliance on Linear Interpolation and Backward-Filling to create a complete time series introduces the risk of model bias, as it assumes steady or static rates of change which may not reflect reality. Furthermore, the model's predictive strength for the observed 2018 values does not guarantee the long-term accuracy of the 2025 forecast, which was based on linear extrapolation of the independent variables' trends. Finally, the high reliance on Country Dummies (55% feature importance) implies that significant underlying, unobserved country-specific factors (e.g., policy, geography, stigma) are driving outcomes, which limits the prescriptive utility of the numerical features.

8.4 Future Work

Future work should focus on three primary areas to extend this research. First, new models should incorporate **Integration of Granular Data**, including social and structural determinants like stigma metrics, health expenditure per capita, and specific intervention coverage rates for key populations. Second, researchers should explore **Alternative Forecasting Models**, comparing the Random Forest Regressor (RFR) with deep learning time series models like **Long Short-Term Memory (LSTM)** or **Prophet**, which are better equipped to model temporal dependencies and non-linear trends without relying on manual feature extrapolation. Third, the analysis should transition from correlation and forecasting to **Causal Inference** modeling to determine the **true causal impact** of Antiretroviral Therapy (ART) and Prevention of Mother-to-Child Transmission (PMTCT) on prevalence and mortality, yielding more robust and actionable policy recommendations.

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